

Likelihood-free phylogenetics



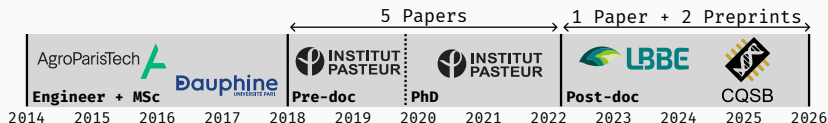
Luc Blassel

Auditions CNRS 51/02

May 20th 2026

slides: lucblassel.com/CNRS_2026.pdf

Personal research context & history



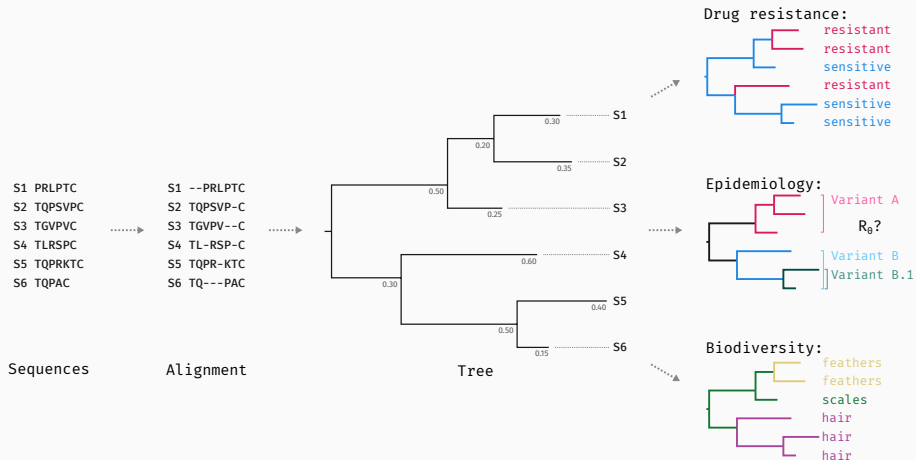
An interdisciplinary setting:

- *Algorithmics & theory*: **Improving** sequence **alignments**
- *Data analysis*: **Learning** from alignments
- *Methods & engineering*: **Deep Learning** for phylogenetic **inference**

General research framework:

Methods to gather **evolutionary** insight from **sequences**

General research context - Phylogenetic analysis



Previous research work

Previous works - Improving sequence alignment

PhD

keywords: Sequence Bioinformatics, combinatorics

Reference ...MTATQPRLPACVPVCMVILTIPRSTCMLIV...

Reads
TQ---PAC
PRLPTC
TGVVPC
TQ-PVC
TL-RSPC
TQPRKTC

S1 PRLPTC
S2 TQPSVPC
S3 TGVVPC
S4 TLRSPC
S5 TQPRKTC
S6 TQPAC

S1 --PRLPTC
S2 TQPSVP-C
S3 TGVVPV--C
S4 TL-RSP-C
S5 TQPR-KTC
S6 TQ---PAC

Sequences

Alignment

Tree

Drug resistance:

resistant
resistant
sensitive
resistant
sensitive
sensitive

Epidemiology:

Variant A
R₀?
Variant B
Variant B.1

Biodiversity:

feathers
feathers
scales
hair
hair
hair

Blassel et al. 2022, RECOMB-SEQ;

keywords: Sequence Bioinformatics, combinatorics

Reference ...MTATQPRLPACVPVCMVILTIPRSTCMLIV...

Reads
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PRLPTC
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TL-RSPC
TQPRKTC

S1 PRLPTC
S2 TQPSVPC
S3 TGVPVC
S4 TLRSPC
S5 TQPRKTC
S6 TQPAC

S1 --PRLPTC
S2 TQPSVP-C
S3 TGVPV--C
S4 TL-RSP-C
S5 TQPR-KTC
S6 TQ---PAC

Sequences

Alignment

Summary:

Goal: improve read-mapping

Method: sequence-to-sequence transformations

Impact: Adapted in a genome assembly tool ¹

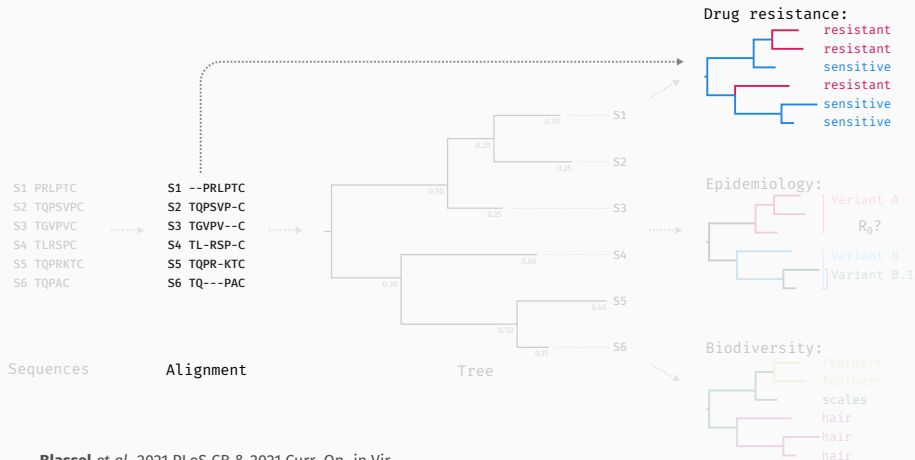
Drug resistance:
resistant

Tree

Biodiversity:

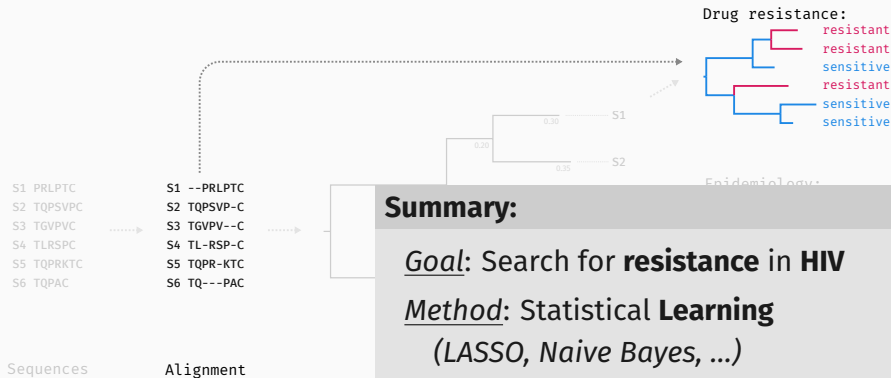
Blassel et al. 2022, RECOMB-SEQ; ¹Faure et al. 2025

keywords: Data Analysis, Pipelining, Machine Learning



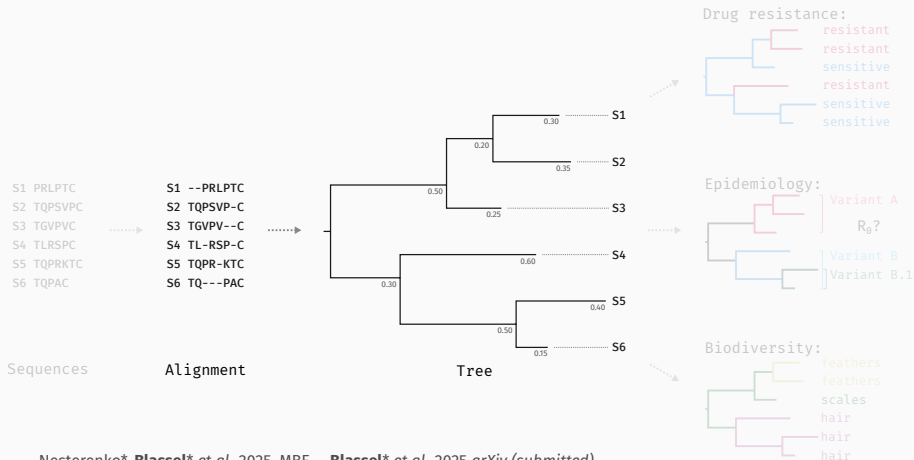
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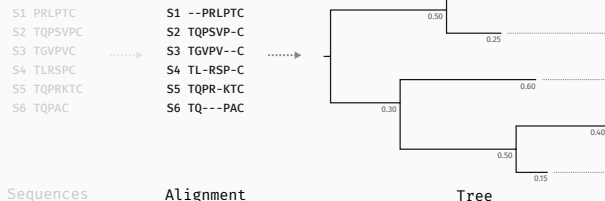
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keywords: Pytorch, Deep Learning,



Nesterenko*, Blassel* et al. 2025, MBE Blassel* et al. 2025 arXiv (submitted)

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Summary:

Goal: Infer
tree distributions
from alignments

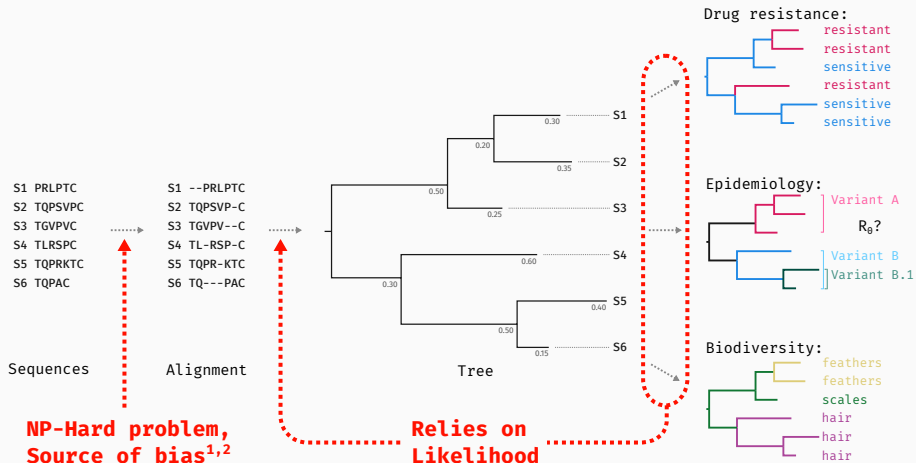
Method: Deep
neural networks
(Phyloformer 1&2)

Impact: Handles
intractable
models

Nesterenko*, Blassel* et al. 2025, MBE Blassel* et al. 2025 arXiv (submitted)

Context - Methodological Barriers

¹Liu et al. 2011, ²Wong et al. 2008



Context - The issue with likelihoods

State of the art relies on **Likelihood** ($\mathcal{L}$):

Max. Likelihood: $\theta^* = \underset{\theta}{\operatorname{argmax}} \mathbf{p}(\mathbf{x}|\theta, \mathbf{M})$

Bayesian: $p(\theta|x, M) \propto \mathbf{p}(\mathbf{x}|\theta, \mathbf{M})p(\theta)p(M)$

Costly^{1,2}

$$\mathcal{L} = \mathcal{O}(ns^3 + nLs^2)$$

Limits model-complexity

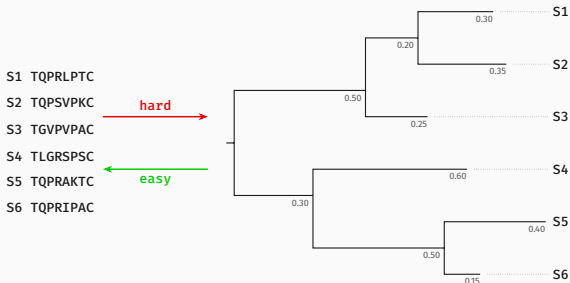
Settings with **impractical** $\mathcal{L}$

- **Potts:** models site **co-evolution**³
- **MTBD:** can model **epidemics** with distinct **populations**⁴

n : nb. of leaves, L : length of sequence, s : size of alphabet

¹Bryant et al. 2005, ²Felsenstein 2004, ³Rodriguez Horta et al. 2019, ⁴Barido-Sottani et al. 2020

Context - Likelihood-free inference

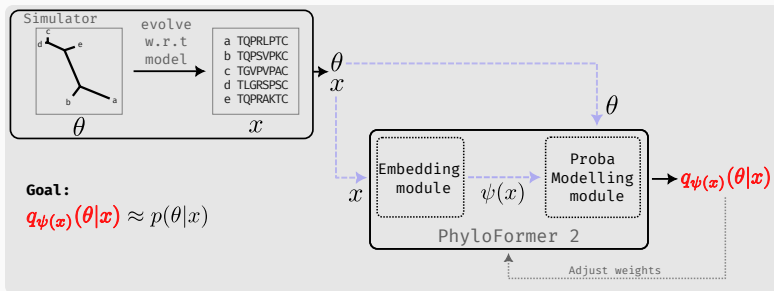


Setting:

- **Hard** to compute likelihood
- **Easy** to **simulate**

Solution:

1. **Simulate** many datapoints under model priors
2. **Learn** $f(x) : x \mapsto \theta$



Training: $\psi^* = \operatorname{argmax}_{\psi} \sum_{i \in \text{train}} q_{\psi(x_i)}(\theta_i|x_i)$ ¹

Inference: we **sample** from $q_{\psi^*(x_E)}(\theta_E|x_E)$

Blassel et al. 2025, *arXiv (submitted)*

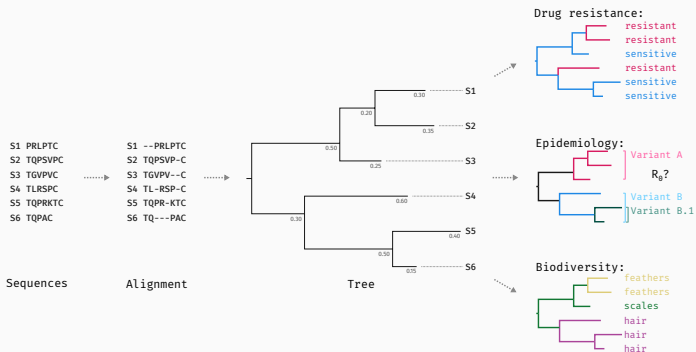
¹ equiv. to: $\operatorname{argmin}_{\psi} D_{\text{KL}}(q_{\psi(x)}(\theta|x) || p(\theta|x))$

Research Project

Project - Overview

Goal:

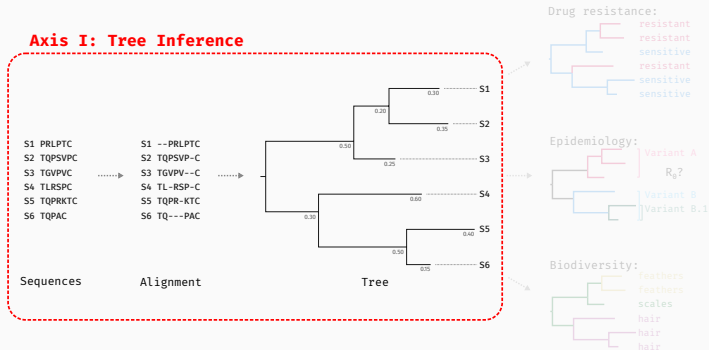
likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



Project - Overview

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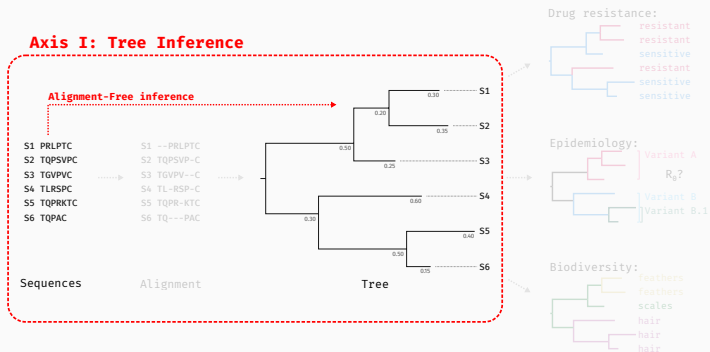
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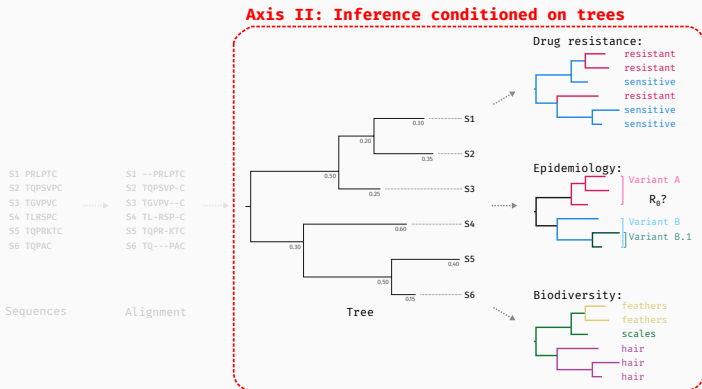
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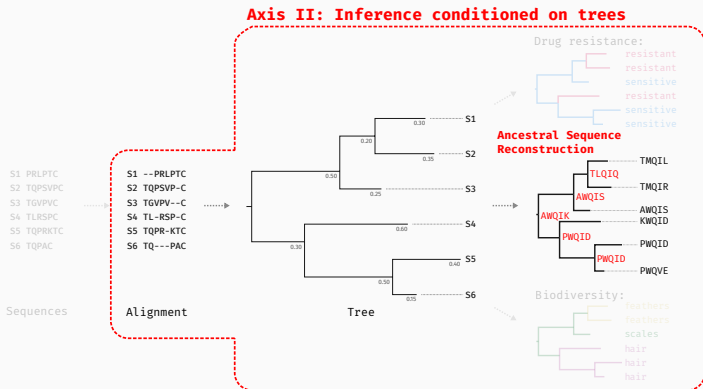
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Project - Overview

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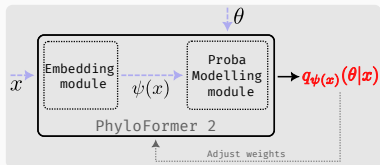
likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



Project - Ax. I - Alignment and likelihood-free inference

Problem (Reminder):

- **Multiple** sequence **alignment** (MSA) is **hard**
- **Uncertainty** in MSA can be an important source of bias^{1,2}



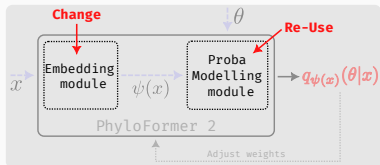
- **Hard** because we **lose** the **structure** induced by **MSAs**

¹Liu et al. 2011, ²Wong et al. 2008

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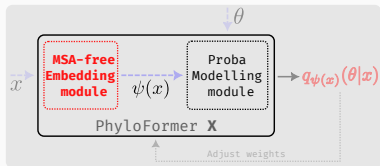
- **Hard** because we **lose** the **structure** induced by **MSAs**
- **Adapt** our existing posterior estimation **framework**

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Project - Ax. I - Alignment and likelihood-free inference

Problem (Reminder):

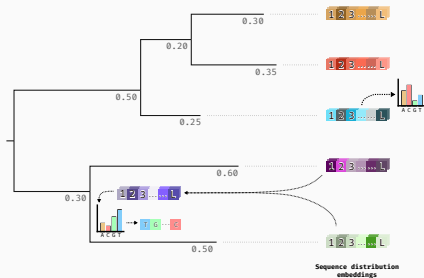
- **Multiple** sequence **alignment** (MSA) is **hard**
- **Uncertainty** in MSA can be an important source of bias^{1,2}



- **Hard** because we **lose** the **structure** induced by **MSAs**
- **Adapt** our existing posterior estimation **framework**
- **Not trivial** and this is only a **first-approach**

¹Liu et al. 2011, ²Wong et al. 2008

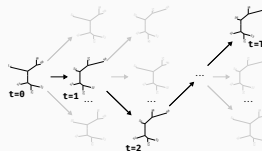
Project - Ax. II - Ancestral sequence reconstruction



- **per-site distribution** over alphabet
- *Training:* we have **simulated ancestral** seq.
- **Flexible** approach for other **branch-specific** parameters: Dates, speciation rates, ...

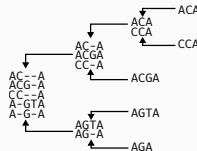
posterior calibration issues:

⇒ More **flexible** posterior modelling with **diffusion**



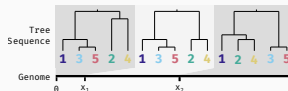
Alignments can still be **useful**:

⇒ **Integrate** alignment-free approach **to MSA** pipeline



Modelling **more complex processes**:

⇒ Infer **tree-sequences** to start tackling **recombination**



Scaling to large number of sequences:

- **17M** SARS-CoV-2 genomes in GISAID
- **900k** genomes in GTDB
- **250k** genomes in IGM database

Extend from **trees** to **networks**:

- **Horizontal transfer** is common in **bacteria**¹
- **Recombination** is frequent in **viruses**²
- *Collaboration with UW-Madison*

numbers: [GISAID](#), [GTDB](#), [IMG](#)

¹Arnold et al. 2022, ²Pérez-Losada et al. 2015

Integration & current collaborations

Analytical Genomics team @ CQSB:

Expertise: SBI/ML (L. Jacob), Evolution (M. Weigt, A. Carbone)

Collaborators: L. Jacob & P. Barrat-Charlaix, M. Weigt (*planned*)

Bioinfo team @ LISN:

Expertise: SBI/ML (F. Pouyet, F. Jay), Trees/Graphs (L. Planche)

Collaborators: F. Pouyet & F. Jay (*planned*)

Modelling of Pathogens team @ IBENS & IP:

Expertise: Evolution & Modelling (A. Zhukova, H. Morlon)

Collaborators: A. Zhukova

Research output:

6 Peer-reviewed **papers** 
incl. 4 as (co)1st author
2 submitted **Preprints**.
Code on `git(hub|lab)`.

Supervision:

L3: 1, M1: 1, M2: 2
Oct 2026: **PhD** co-supervision
with **A. Zhukova**

Reviewing:

Journals: MBE, Syst. Bio, PLoS
CB, Bioinformatics
Conferences: RECOMB-SEQ,
ICLR, ICML

Collaborations:

France: CIRB, IBENS, Institut
Pasteur, LBBE
International: Penn. State,
EPFL, UW-Madison

Funded by:

PhD: INCEPTION & PRAIRIE
PostDoc: ANR DEELOGENY

**Thank you
for listening!**

References

- Arnold, B. J. et al. (2022). **Horizontal Gene Transfer and Adaptive Evolution in Bacteria.** In: *Nature Reviews Microbiology* 20.4, pp. 206–218.
- Barido-Sottani, J. et al. (2020). **A Multitype Birth–Death Model for Bayesian Inference of Lineage-Specific Birth and Death Rates.** In: *Syst Biol* 69.5, pp. 973–986.
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- Liu, K. et al. (2011). **Multiple Sequence Alignment: A Major Challenge to Large-Scale Phylogenetics.** In: *PLoS Curr* 2, RRN1198.
- Pérez-Losada, M. et al. (2015). **Recombination in Viruses: Mechanisms, Methods of Study, and Evolutionary Consequences.** In: *Infect Genet Evol* 30, pp. 296–307.
- Rodriguez Horta, E. et al. (2019). **Toward Inferring Potts Models for Phylogenetically Correlated Sequence Data.** In: *Entropy* 21.11, p. 1090.
- Wong, K. M. et al. (2008). **Alignment Uncertainty and Genomic Analysis.** In: *Science* 319.5862, pp. 473–476.

This is where the TOC will be

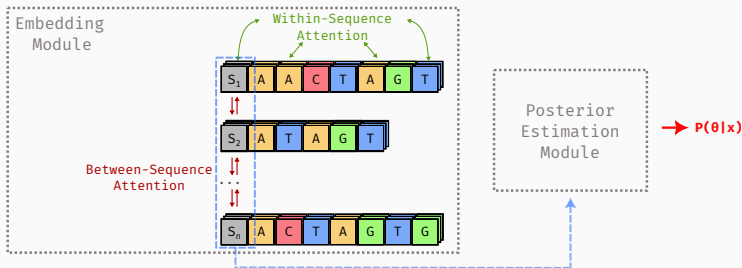
Published Articles:

- Nesterenko, L. et al. (2025). **Phyloformer: Fast, Accurate, and Versatile Phylogenetic Reconstruction with Deep Neural Networks**. In: *Molecular Biology and Evolution* 42.4, msaf051.
- Blassel, L., P. Medvedev, et al. (2022). **Mapping-Friendly Sequence Reductions: Going beyond Homopolymer Compression**. In: *iScience*, p. 105305.
- Blassel, L., A. Tostevin, et al. (2021). **Using Machine Learning and Big Data to Explore the Drug Resistance Landscape in HIV**. In: *PLOS Computational Biology* 17.8, e1008873.
- Blassel, L., A. Zhukova, et al. (2021). **Drug Resistance Mutations in HIV: New Bioinformatics Approaches and Challenges**. In: *Current Opinion in Virology* 51, pp. 56–64.
- Zhukova, A. et al. (2021). **Origin, Evolution and Global Spread of SARS-CoV-2**. In: *Comptes Rendus. Biologies* 344.1, pp. 57–75.
- Lemoine, F. et al. (2020). **COVID-Align: Accurate Online Alignment of hCoV-19 Genomes Using a Profile HMM**. In: *Bioinformatics* 37.12, pp. 1761–1762.

Submitted Preprints:

- Garot, V. et al. (2026). **Teddy: Neural Inference of Epidemiological Parameters from Viral Sequences**.
- Blassel, L., B. Boussau, et al. (2025). **Likelihood-Free Inference of Phylogenetic Tree Posterior Distributions**.

Project - Alignment-free embedding



- **Re-use** existing posterior estimation **module**
- **Easy** to create a first **prototype**
- *Challenges*: overcome **information bottlenecks** & adaptation to **empirical data**

Project - Assembly-free phylogenetic inference

Schema

Project - Tree-sequence inference

frame

Project - Inference on trees, GNNs for R_0

frame

Project - Joint tree-parameter inference, dating

env